Double-Angle Formulas and Trigonometric Equations

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Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

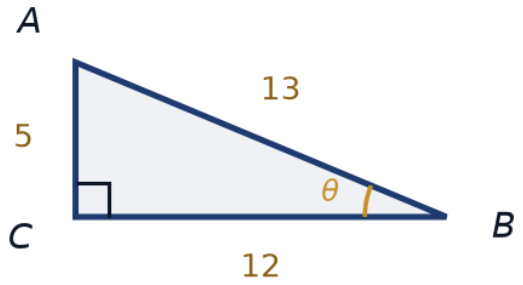
1 [Foundational] • Target: Double-angle formulas

Given that $\cos\theta = -\frac{3}{5}$ and θ lies in Quadrant II, find:

a $\sin 2\theta$ **b** $\cos 2\theta$ **c** $\tan 2\theta$

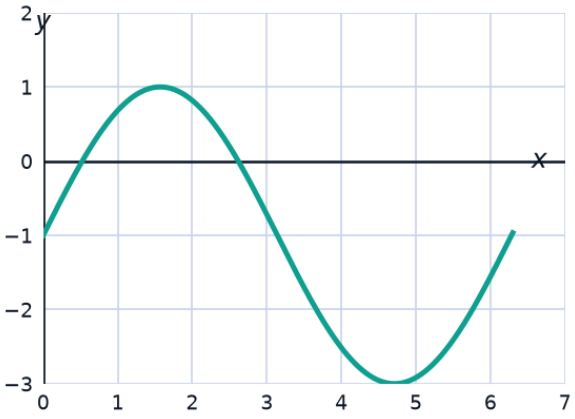
2 [Foundational] • Target: Double-angle formulas

Use a double-angle formula to write $\cos^2 x$ in terms of $\cos 2x$, and state the corresponding formula for $\sin^2 x$.



3 [Foundational] • Target: Solving trigonometric equations

Solve $2\sin x - 1 = 0$ on the interval $[0, 2\pi)$.



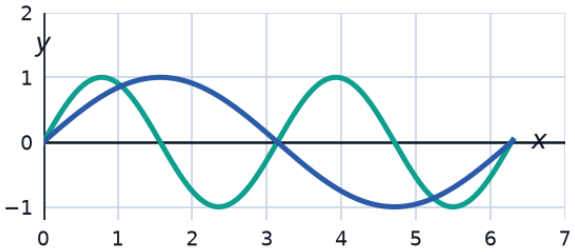
4 [Foundational] • Target: Solving trigonometric equations

Solve $2\cos^2 x - \cos x - 1 = 0$ on the interval $[0, 2\pi)$.

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5 [Foundational] • Target: Solving trigonometric equations

Solve $\sin 2x = \sin x$ on the interval $[0, 2\pi)$.



6 [Foundational] • Target: Solving trigonometric equations

Solve $\tan^2 x = 3$ on the interval $[0, 2\pi)$.

7 [Foundational] • Target: Half-angle formulas

Use a half-angle formula to find the exact value of $\cos 15°$ (using $15° = \frac{30°}{2}$).

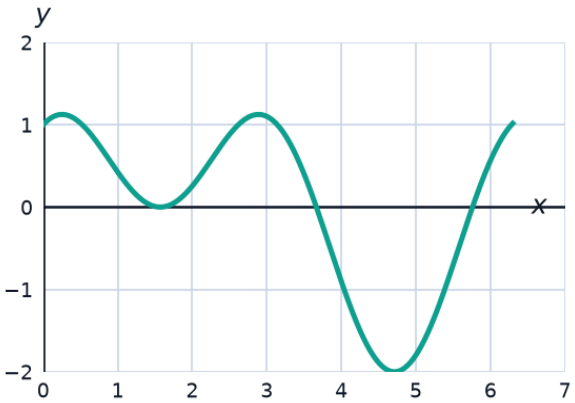
Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Half-angle formulas

Given $\cos\theta = \frac{7}{25}$ with $0 < \theta < \frac{\pi}{2}$, find $\sin\frac{\theta}{2}$ exactly.

9 [Intermediate] • Target: Equations requiring double-angle substitution

Solve $\cos 2x + \sin x = 0$ on the interval $[0, 2\pi)$, using $\cos 2x = 1 - 2\sin^2 x$.



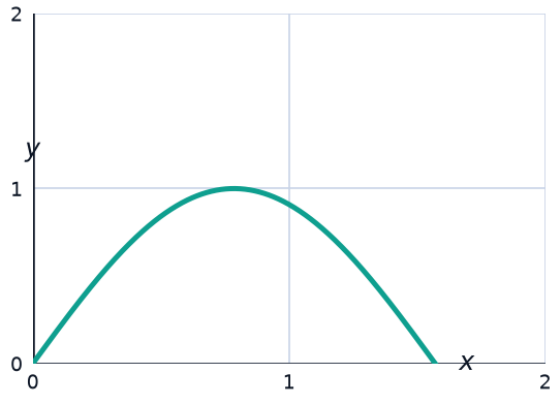
10 [Intermediate] • Target: Equations requiring double-angle substitution

Solve $\sin 2x - \cos x = 0$ on the interval $[0, 2\pi)$.

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11 [Intermediate] • Target: Applications of double-angle formulas

A projectile's range is given by $R(\theta) = \frac{v^2 \sin 2\theta}{g}$. Using the double-angle formula, explain why the maximum range occurs at $\theta = 45°$.



12 [Intermediate] • Target: Applications of double-angle formulas

Given $\sin\theta = \frac{1}{3}$ with θ in Quadrant I, find $\cos 2\theta$ and $\sin 2\theta$ exactly.

13 [Intermediate] • Target: More equation practice

Solve $4\sin^2 x - 3 = 0$ on the interval $[0, 2\pi)$.

14 [Intermediate] • Target: More equation practice

Solve $\cos 2x = \cos x$ on the interval $[0, 2\pi)$, using $\cos 2x = 2\cos^2 x - 1$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Half-angle formulas for tangent • Cross-Lesson: 8.03 Introduction to Limits

State the half-angle formula for $\tan\frac{\theta}{2}$ in terms of $\sin\theta$ and $\cos\theta$ (the form avoiding a square root), and use it to find $\tan 15°$.

16 [Challenging] • Target: Equations with multiple angle types • Cross-Lesson: 9.01 Arithmetic Sequences and Series

Solve $\sin 2x + \sin x = 0$ on $[0, 2\pi)$.

17 [Challenging] • Target: Equations with multiple angle types • Cross-Lesson: 9.02 Geometric Sequences and Series

Solve $\cos 2x = 1 - 3\sin x$ on $[0, 2\pi)$, using $\cos 2x = 1 - 2\sin^2 x$.

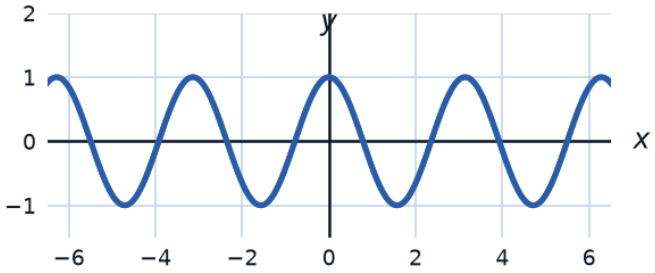
18 [Challenging] • Target: Triple-angle preview • Cross-Lesson: 9.03 The Binomial Theorem

Using $\sin(2x + x) = \sin 2x\cos x + \cos 2x\sin x$ together with the double-angle formulas, derive the triple-angle formula $\sin 3x = 3\sin x - 4\sin^3 x$.

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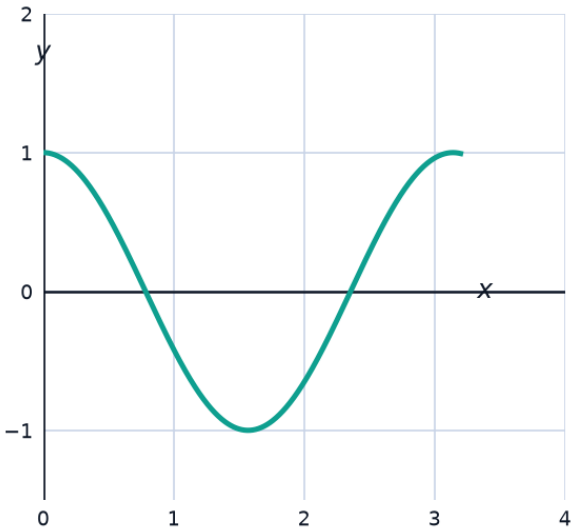
19 [Challenging] • Target: Graphical verification of a double-angle identity • Cross-Lesson: 1.01 Functions and Function Notation

The functions $y = \cos 2x$ and $y = 1 - 2\sin^2 x$ are graphed together below. Explain what it means that the two curves appear identical, and state the identity this confirms.



20 [Challenging] • Target: Solving equations restricted to a smaller interval • Cross-Lesson: 1.02 Domain and Range

Solve $2\cos 2x - 1 = 0$ for $x \in [0, \pi)$ only (note the restricted domain).



Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Double-Angle Formulas

State the double-angle formula for $\sin(2\theta)$.

a. $2\sin\theta\cos\theta$

b. $\cos^2\theta - \sin^2\theta$

c. $2\sin\theta$

d. $\sin^2\theta$

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22 [Foundational] • Target: Double-Angle Formulas

Which of the following is NOT an equivalent form of $\cos(2\theta)$?

a. $1 - 2\cos^2\theta$

b. $\cos^2\theta - \sin^2\theta$

c. $2\cos^2\theta - 1$

d. $1 - 2\sin^2\theta$

23 [Foundational] • Target: Double-Angle Formulas

If $\cos\theta = 4/5$ with θ in Quadrant IV, find $\sin(2\theta)$.

a. $-24/25$

b. $24/25$

c. $-7/25$

d. $7/25$

24 [Foundational] • Target: Half-Angle and Power-Reducing Formulas

State the power-reducing formula for $\sin^2 x$.

a. $\frac{1-\cos(2x)}{2}$

b. $\frac{1+\cos(2x)}{2}$

c. $\frac{1-\sin(2x)}{2}$

d. $1 - \cos^2 x$

25 [Foundational] • Target: Half-Angle and Power-Reducing Formulas

Find the exact value of $\cos(22.5^\circ)$ using the half-angle formula $\cos(\theta/2) = \sqrt{\frac{1+\cos\theta}{2}}$.

a. $\frac{\sqrt{2+\sqrt{2}}}{2}$

b. $\frac{\sqrt{2-\sqrt{2}}}{2}$

c. $\frac{\sqrt{2}+1}{2}$

d. $\frac{\sqrt{3}+1}{2}$

26 [Foundational] • Target: Half-Angle and Power-Reducing Formulas

Rewrite $\cos^4 x$ in terms of first powers of cosines of multiple angles.

a. $\frac{3}{8} + \frac{1}{2}\cos(2x) + \frac{1}{8}\cos(4x)$

b. $\frac{1}{4}(1 + \cos(2x))^2$

c. $\frac{1}{2} + \frac{1}{2}\cos(4x)$

d. $\frac{3}{8} - \frac{1}{2}\cos(2x)$

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27 [Foundational] • Target: Solving Linear Trigonometric Equations

Solve $\sqrt{3}\tan x + 1 = 0$ on $[0, 2\pi)$.

a. $\frac{5\pi}{6}$ and $\frac{11\pi}{6}$

b. $\frac{2\pi}{3}$ and $\frac{5\pi}{3}$

c. $\frac{\pi}{6}$ and $\frac{7\pi}{6}$

d. $\frac{5\pi}{6}$ and $\frac{7\pi}{6}$

28 [Foundational] • Target: Solving Linear Trigonometric Equations

Find all solutions to $\cos(2x) = 0$ on $[0, \pi)$.

a. $\frac{\pi}{4}$ and $\frac{3\pi}{4}$

b. $\frac{\pi}{2}$

c. $\frac{\pi}{4}$ only

d. $\frac{\pi}{2}$ and $\frac{3\pi}{2}$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Quadratic-Type and Multiple-Angle Equations

Solve $2\sin^2 x - \sin x - 1 = 0$ on $[0, 2\pi)$.

a. $\frac{\pi}{2}, \frac{7\pi}{6}, \frac{11\pi}{6}$

b. $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{3\pi}{2}$

c. $\frac{\pi}{2}, \frac{5\pi}{6}$

d. $\frac{\pi}{3}, \frac{5\pi}{3}$

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30 [Intermediate] • Target: Quadratic-Type and Multiple-Angle Equations

Solve $\sin(2x) = \cos x$ on $[0, 2\pi)$.

a. $\frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}, \frac{3\pi}{2}$

b. $\frac{\pi}{6}, \frac{5\pi}{6}$

c. $\frac{\pi}{3}, \frac{2\pi}{3}$

d. $\frac{\pi}{2}, \frac{3\pi}{2}$

31 [Intermediate] • Target: Quadratic-Type and Multiple-Angle Equations

Solve $\cos(2x) + \cos x = 0$ on $[0, 2\pi)$.

a. $\frac{\pi}{3}, \pi, \frac{5\pi}{3}$

b. $\frac{\pi}{6}, \pi, \frac{11\pi}{6}$

c. $\frac{\pi}{2}, \frac{3\pi}{2}$

d. $\frac{\pi}{3}, \frac{2\pi}{3}$

32 [Intermediate] • Target: Double-Angle Formulas

If $\cos\theta = -\frac{7}{25}$ and $\frac{\theta}{2}$ is in Quadrant I, find $\sin\left(\frac{\theta}{2}\right)$ using the half-angle formula.

a. $\frac{16}{25}$

b. $-\frac{4}{5}$

c. $\frac{4}{5}$

d. $\frac{9}{10}$

33 [Intermediate] • Target: Half-Angle and Power-Reducing Formulas

If $\cos\theta = \frac{23}{25}$ and $\frac{\theta}{2}$ is in Quadrant I, find $\sin\left(\frac{\theta}{2}\right)$ using the half-angle formula.

a. $-\frac{1}{5}$

b. $\frac{3}{10}$

c. $\frac{1}{5}$

d. $\frac{1}{25}$

34 [Intermediate] • Target: Solving Linear Trigonometric Equations

If $\cos\theta = \frac{17}{25}$ and $\frac{\theta}{2}$ is in Quadrant I, find $\sin\left(\frac{\theta}{2}\right)$ using the half-angle formula.

a. $\frac{1}{2}$

b. $-\frac{2}{5}$

c. $\frac{2}{5}$

d. $\frac{4}{25}$

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35 [Intermediate] • Target: Quadratic-Type and Multiple-Angle Equations

If $\cos\theta = \frac{7}{25}$ and $\frac{\theta}{2}$ is in Quadrant I, find $\sin\left(\frac{\theta}{2}\right)$ using the half-angle formula.

a. $\frac{9}{25}$

b. $\frac{3}{5}$

c. $-\frac{3}{5}$

d. $\frac{7}{10}$

36 [Intermediate] • Target: Double-angle formulas

If $\cos\theta = -\frac{1}{8}$ and $\frac{\theta}{2}$ is in Quadrant I, find $\sin\left(\frac{\theta}{2}\right)$ using the half-angle formula.

a. $-\frac{3}{4}$

b. $\frac{17}{20}$

c. $\frac{9}{16}$

d. $\frac{3}{4}$

37 [Intermediate] • Target: Solving trigonometric equations

If $\cos\theta = \frac{1}{2}$ and $\frac{\theta}{2}$ is in Quadrant I, find $\sin\left(\frac{\theta}{2}\right)$ using the half-angle formula.

a. $\frac{1}{4}$

b. $\frac{1}{2}$

c. $\frac{3}{5}$

d. $-\frac{1}{2}$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Half-angle formulas • Cross-Lesson: 9.03 The Binomial Theorem

If $\cos\theta = \frac{7}{9}$ and $\frac{\theta}{2}$ is in Quadrant I, find $\sin\left(\frac{\theta}{2}\right)$ using the half-angle formula.

a. $\frac{13}{30}$

b. $\frac{1}{3}$

c. $-\frac{1}{3}$

d. $\frac{1}{9}$

39 [Challenging] • Target: Equations requiring double-angle substitution • Cross-Lesson: 1.01 Functions and Function Notation

If $\cos\theta = \frac{1}{9}$ and $\frac{\theta}{2}$ is in Quadrant I, find $\sin\left(\frac{\theta}{2}\right)$ using the half-angle formula.

a. $\frac{2}{3}$

b. $\frac{4}{9}$

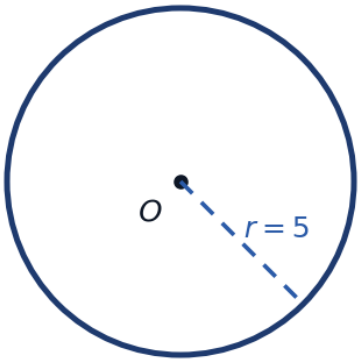
c. $\frac{23}{30}$

d. $-\frac{2}{3}$

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40 [Challenging] • Target: Applications of double-angle formulas • Cross-Lesson: 1.02 Domain and Range

A circle is centered at the origin and passes through the point $P(a, 4)$ in the first quadrant. The equation of a tangent line to the circle at point P is given by $3x + 4y = k$. Find the value of k.



a. 25

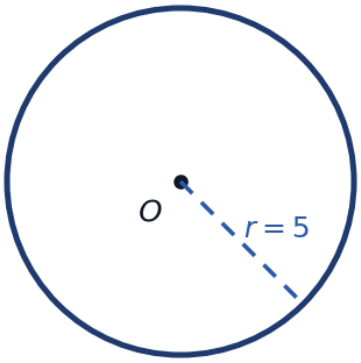
b. 30

c. 35

d. 40

41 [Challenging] • Target: More equation practice • Cross-Lesson: 1.03 Average Rate of Change

On the unit circle, a point P is located at coordinates (x, y). If the angle formed by the positive x-axis and the line segment from the origin to P is θ, what is the value of x in terms of θ?



a. $\cos\theta$

b. $\sin\theta$

c. $\tan\theta$

d. 1

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42 [Challenging] • Target: Half-angle formulas for tangent • Cross-Lesson: 2.01 Right-Triangle Trigonometry

Let the complex number z be represented in polar form as $z = r(\cos\theta + i\sin\theta)$. Suppose z satisfies the equation $z^3 = -64i$. If z is restricted to the fourth quadrant (i.e., $\frac{3\pi}{2} < \arg(z) < 2\pi$), find the real part of z.

a. 4

b. -4

c. $2\sqrt{3}$

d. $-2\sqrt{3}$

43 [Challenging] • Target: Equations with multiple angle types • Cross-Lesson: 2.02 Radian Measure and Arc Length

Solve $\cos(2x) + \sin x = 0$ on $[0, 2\pi)$. (attempt 10)

a. $x = \frac{\pi}{2}, \frac{7\pi}{6}, \frac{11\pi}{6}$

b. $x = \frac{\pi}{2}$ only

c. $x = \frac{7\pi}{6}, \frac{11\pi}{6}$ only

d. $x = 0, \pi$

44 [Challenging] • Target: Triple-angle preview • Cross-Lesson: 2.03 Graphs of Sine and Cosine Functions

Solve $\cos(2x) + \sin x = 0$ on $[0, 2\pi)$. (attempt 4)

a. $x = 0, \pi$

b. $x = \frac{\pi}{2}, \frac{7\pi}{6}, \frac{11\pi}{6}$

c. $x = \frac{7\pi}{6}, \frac{11\pi}{6}$ only

d. $x = \frac{\pi}{2}$ only

45 [Challenging] • Target: Graphical verification of a double-angle identity • Cross-Lesson: 2.04 Law of Sines and Law of Cosines

Solve $\cos(2x) + \sin x = 0$ on $[0, 2\pi)$. (attempt 5)

a. $x = \frac{7\pi}{6}, \frac{11\pi}{6}$ only

b. $x = \frac{\pi}{2}, \frac{7\pi}{6}, \frac{11\pi}{6}$

c. $x = 0, \pi$

d. $x = \frac{\pi}{2}$ only

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